

8 operating specifications

8.1 absolute maximum ratings

table 8-1 absolute maximum ratings

parameter	absolute maximum rating ^a
ambient storage temperature	-40°C to +95°C
supply voltage (with respect to ground) ^b	V_{DD-A} 4.5V
	V_{DD-D} 3V
	V_{DD-IO} 4.5V
electro-static discharge (ESD)	human body model 2000V
	machine model 200V
all input/output voltages (with respect to ground)	-0.3V to $V_{DD-IO} + 1V$
I/O current on any input or output pin	±200 mA
peak solder temperature (10 second dwell time)	245°C

- a. exceeding the absolute maximum ratings shown above invalidates all AC and DC electrical specifications and may result in permanent damage to the device. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.
- b. for negative voltage with respect to ground, V_{DD-A} (-4.5V), V_{DD-C} (-3V), V_{DD-IO} (-4.5V)

8.2 functional temperature

table 8-2 functional temperature

parameter	range
operating temperature ^a	-30°C to +70°C junction temperature
stable image temperature ^b	0°C to +50°C junction temperature

- a. sensor functions but image quality may be noticeably different at temperatures outside of stable image range
- b. image quality remains stable throughout this temperature range

8.3 DC characteristics

table 8-3 DC characteristics ($-30^{\circ}\text{C} < T_J < 70^{\circ}\text{C}$) (sheet 1 of 2)

symbol	parameter	min	typ	max	unit
power supply					
V_{DD-A}	supply voltage (analog)	2.6	2.8	3.0	V
V_{DD-D}^a	supply voltage (digital core)	1.425	1.5	1.575	V
V_{DD-IO}	supply voltage (digital I/O)	1.71	1.8	3.0	V
internal DVDD short to DVDD, DVP output, AVDD = 2.8V, DOVDD = 2.8V					
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	2592 x 1944 @ 15 fps JPG		110	140	mA
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	1080p @ 30 fps JPG		100	130	mA
I_{DD-A}	operating current		32	42	mA
I_{DD-DO}	720p @ 60 fps		100	42	mA
I_{DD-A}	operating current		32	40	mA
I_{DD-DO}	720 @ 30 fps YUV		58	72	mA
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	VGA @ 30 fps		58	72	mA
internal DVDD, EVDD short to DVDD, MIPI output, AVDD = 2.8V, DOVDD = 1.8V					
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	2592 x 1944 @ 15 fps JPG		110	140	mA
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	2592 x 1944 @ 15 fps YUV		100	130	mA
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	1080p @ 30 fps JPG		100	130	mA
I_{DD-A}	operating current		30	40	mA
I_{DD-DO}	1080p @ 30 fps YUV		90	115	mA
I_{DD-A}	operating current		32	42	mA
I_{DD-DO}	720 @ 30 fps YUV		54	70	mA
external DVDD, EVDD short to DVDD, DVP output, AVDD = 2.8V, DOVDD = 2.8V					
I_{DD-A}	operating current		30	40	mA
I_{DD-D}	operating current		98	125	mA
I_{DD-DO}	2592 x 1944 @ 15 fps JPG		9	12	mA

table 8-3 DC characteristics ($-30^{\circ}\text{C} < T_J < 70^{\circ}\text{C}$) (sheet 2 of 2)

symbol	parameter	min	typ	max	unit
standby current					
I _{DDS-SCCB}	standby current ^b		20	50	μA
I _{DDS-PWDN}			20	50	μA
digital inputs (typical conditions: AVDD = 2.8V, DVDD = 1.5V, DOVDD = 1.8V)					
V _{IL}	input voltage LOW			0.54	V
V _{IH}	input voltage HIGH	1.26			V
C _{IN}	input capacitor			10	pF
digital outputs (standard loading 25 pF)					
V _{OH}	output voltage HIGH	1.62			V
V _{OL}	output voltage LOW			0.18	V
serial interface inputs ^c					
V _{IL}	SIOC and SIOD	-0.5	0	0.54	V
V _{IH}	SIOC and SIOD	1.26	1.8	3.0	V

- a. using the internal DVDD regulator is strongly recommended for minimum power down current
- b. standby current is based on room temperature
- c. based on $\text{DOVDD} = 1.8\text{V}$.

8.4 AC characteristics

table 8-4 AC characteristics ($T_A = 25^\circ\text{C}$, $V_{DD-A} = 2.8\text{V}$)

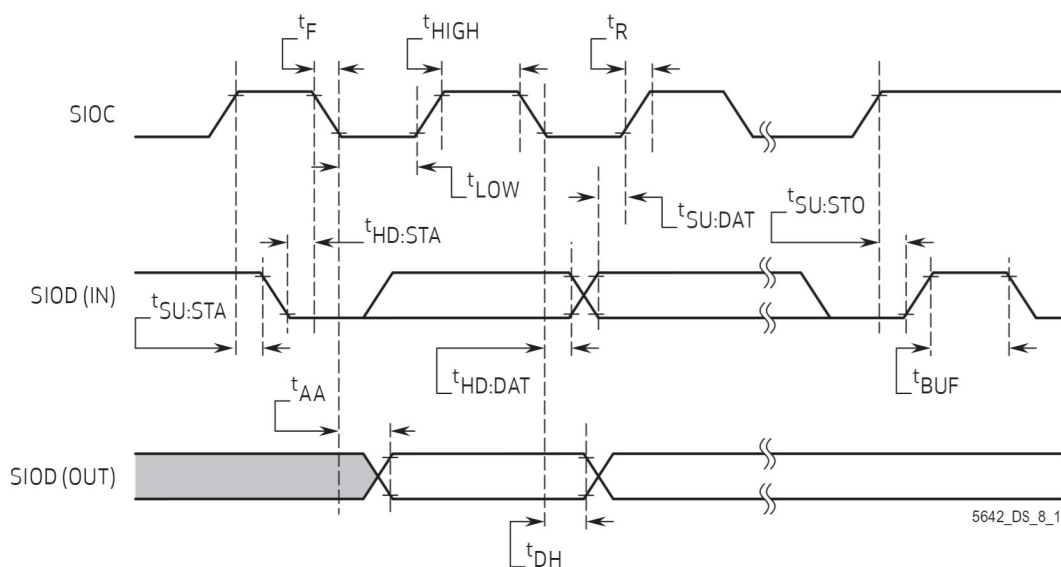
symbol	parameter	min	typ	max	unit
ADC parameters					
B	analog bandwidth		30		MHz
DLE	DC differential linearity error		0.5		LSB
ILE	DC integral linearity error		1		LSB
	settling time for hardware reset			<1	ms
	settling time for software reset			<1	ms
	settling time for resolution mode change			<1	ms
	settling time for register setting			<300	ms

table 8-5 timing characteristics

symbol	parameter	min	typ	max	unit
oscillator and clock input					
f_{OSC}	frequency (XVCLK) ^a	6	24	54	MHz
t_r, t_f	clock input rise/fall time ^b			5 (10 ^c)	ns
f_{PCLK}	parallel port output pixel clock		48 ^d	96 ^e	MHz

- for input clock range 6~27 MHz, the OV5640 can tolerate input clock jitter up to 1ns, for input clock range to 54MHz, the OV5640 can tolerate input clock jitter up to 500 ps
- if the PLL is bypassed, the delay from input clock to output clock is approximately 4~5 ns
- if using the internal PLL
- typical PCLK is 48 MHz when sensor output is smaller size (VGA YUV or below) or full size compression
- 96 MHz is for sensor RAW data output at 15 fps or YUV output at 7.5 fps. For higher speeds such as 5 megapixel YUV @ 15 fps, OmniVision recommends using the MIPI two-lane interface.

figure 8-1 SCCB interface timing

table 8-6 SCCB interface timing specifications^a

symbol	parameter	min	typ	max	unit
f_{SIOC}	clock frequency			400	KHz
t_{LOW}	clock low period	1.3			μs
t_{HIGH}	clock high period	0.6			μs
t_{AA}	SIOC low to data out valid	0.1		0.9	μs
t_{BUF}	bus free time before new start	1.3			μs
$t_{\text{HD:STA}}$	start condition hold time	0.6			μs
$t_{\text{SU:STA}}$	start condition setup time	0.6			μs
$t_{\text{HD:DAT}}$	data in hold time	0			μs
$t_{\text{SU:DAT}}$	data in setup time	0.1			μs
$t_{\text{SU:STO}}$	stop condition setup time	0.6			μs
$t_{\text{R}}, t_{\text{F}}$	SCCB rise/fall times			0.3	μs
t_{DH}	data out hold time	0.05			μs

a. SCCB timing is based on 400KHz mode